UKAPA7 Layer U Partial B-Side Offset Report

This report repeats the UKAPA7 comparison workflow for the U layer using:

- UK factory measurements in `UKAPA7U.json`
- Chicago factory measurements in
  `data/tension_summaries/tension_summary_UKAPA7_U.csv`

Residual definition throughout: `Chicago - UK`.

Data Availability

- The UK JSON contains both side A and side B measurements.
- The Chicago summary CSV now contains all currently available U-layer data on
  both sides.
- The available Chicago side-B wires span wire numbers `241` through `967`.
- The current CSV contains `641` non-null side-B measurements and `385` aligned
  side-A comparison rows.

This report still focuses on the available side-B subset, since the U-layer
indexing question being tested here is a B-side offset model.

Working Assumption

For this U-layer analysis, the working assumption is:

- there is no side-B reversal
- there may be a constant wire-number offset

So the tested model family is a simple shift:

`Chicago B wire w` -> `UK B wire w + k`

with no reversal term.

Executive Summary

- Side A data are now available in the Chicago summary and are included below as
  a direct raw-data comparison.
- For the currently available A-side rows, the mean residual is about
  `-0.544 N`.
- Under the current indexing, the partial B-side comparison is already strongly

negative on average, with mean residual about `-0.871 N`.
- Scanning constant shifts from `-80` to `+80` shows the best correlation at shift `-7`.
- That shift improves correlation from `0.509` to `0.586` and reduces the residual width from `0.548 N` to `0.492 N`.
- The improvement is real but not transformative. The Chicago and UK traces remain substantially offset even after the best tested small shift.

Source Artifacts

- Raw A comparison: `data/tension_plots/tension_raw_A_UKAPA7_U.png`
- Raw B comparison, current indexing: `data/tension_plots/tension_raw_B_baseline_UKAPA7_U.png`
- Raw B comparison, best offset: `data/tension_plots/tension_raw_B_shifted_UKAPA7_U.png`
- Residual comparison, baseline vs best offset: `data/tension_plots/tension_residual_B_offset_comparison_UKAPA7_U.png`
- Offset scan: `data/tension_plots/tension_shift_scan_B_UKAPA7_U.png`
- Aligned comparison table: `data/tension_summaries/tension_B_offset_comparison_UKAPA7_U.csv`

Raw A-Side Data

![UKAPA7 U raw A](../data/tension_plots/tension_raw_A_UKAPA7_U.png){ width=100% }

A-Side Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)	Negative Fraction
Side A	385	5.419	5.962	0.104	-0.544	0.524	0.610	0.855

Interpretation

- Side A is now available as a direct UK-versus-Chicago comparison.
- The mean residual is negative, consistent with the same overall direction of change seen elsewhere.
- The A-side correlation is still weak in the currently available subset, so the report keeps its main focus on the B-side offset question rather than trying to build an indexing model for side A.

Raw B-Side Data, Current Indexing

![UKAPA7 U raw B baseline](../data/tension_plots/tension_raw_B_baseline_UKAPA7_U.png){ width=100% }

Current-Indexing Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)
Side B baseline	641	5.331	6.202	0.509	-0.871	0.548	0.895

Raw B-Side Data, Best Constant Offset

![UKAPA7 U raw B shifted](../data/tension_plots/tension_raw_B_shifted_UKAPA7_U.png){ width=100% }

Best-Offset Summary

Comparison	Count	Chicago Mean (N)	UK Mean (N)	Corr	Mean Diff (N)	Std Diff (N)	MAE (N)
Side B shift `-7`	641	5.331	6.200	0.586	-0.869	0.492	0.874

Interpretation

- A simple negative shift of `-7` wires produces the best correlation in the tested range.
- The improvement is visible in both the line plot and the histogram overlap.
- Even after the shift, the Chicago values remain systematically below the UK values by about `0.95 N` on average.

Residual Comparison

This figure compares the residuals for:

- baseline indexing
- the best tested constant offset, shift `-7`

![UKAPA7 U residual comparison](../data/tension_plots/tension_residual_B_offset_comparison_UKAPA7_U.png){ width=100% }

Residual Statistics

Model	Count	Corr	Mean (N)	Median (N)	Std (N)	MAE (N)	RMSE (N)	Negative Fraction
Baseline	641	0.509	-0.871	-0.810	0.548	0.895	1.029	0.972

| Shift `-7` | 641 | 0.586 | -0.869 | -0.756 | 0.492 | 0.874 | 0.999 | 0.989 |

Interpretation

- The best constant shift tightens the residual distribution and increases the correlation.
- The residuals remain strongly negative even after the shift, which means the dominant effect is still a large average Chicago-versus-UK difference.
- Since only part of side B is present in the CSV, this should be interpreted as a partial-window comparison, not a statement about the entire layer.

Offset Scan

![UKAPA7 U shift scan](../data/tension_plots/tension_shift_scan_B_UKAPA7_U.png){ width=100% }

Interpretation

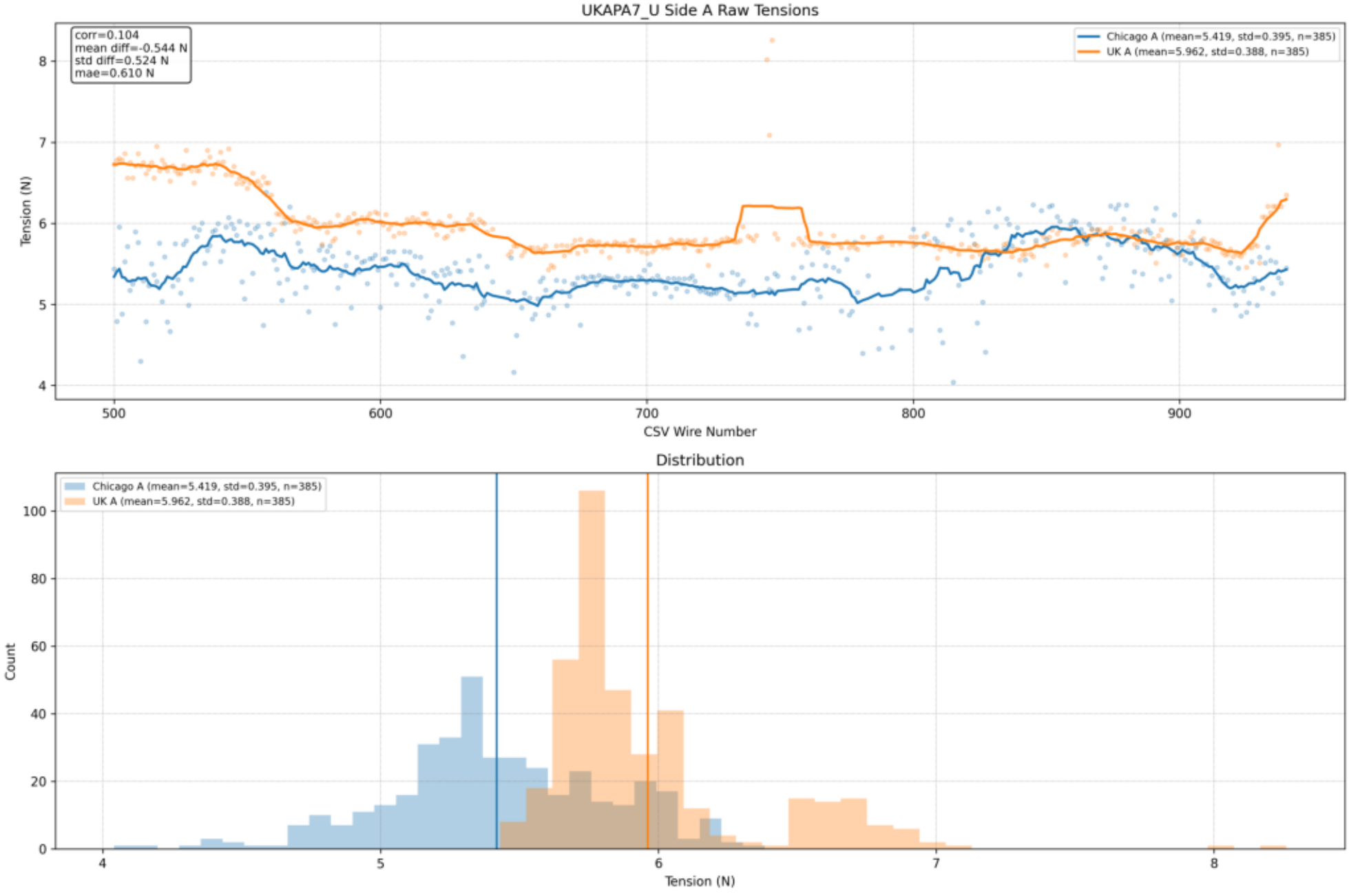
- The correlation curve peaks near shift `-7`.
- The residual standard deviation also reaches a local minimum around the same region.
- Larger negative shifts can lower MAE somewhat, but they do not give the best correlation and are less compelling as a simple indexing-offset explanation.

Conclusion

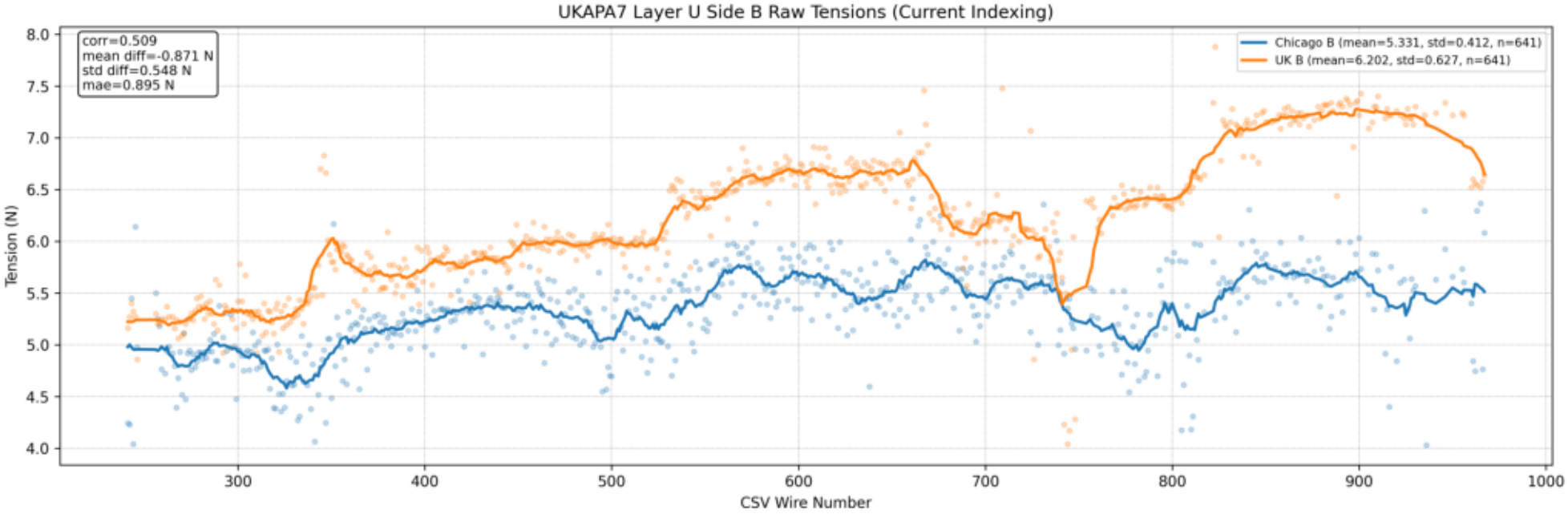
For the available U-layer data, there is no need to invoke a reversal model. The Chicago summary now has partial data on both sides, and the newly available A-side rows are included as a direct raw comparison. Within the side-B window, the best simple index-offset explanation remains a shift of `-7` wires.

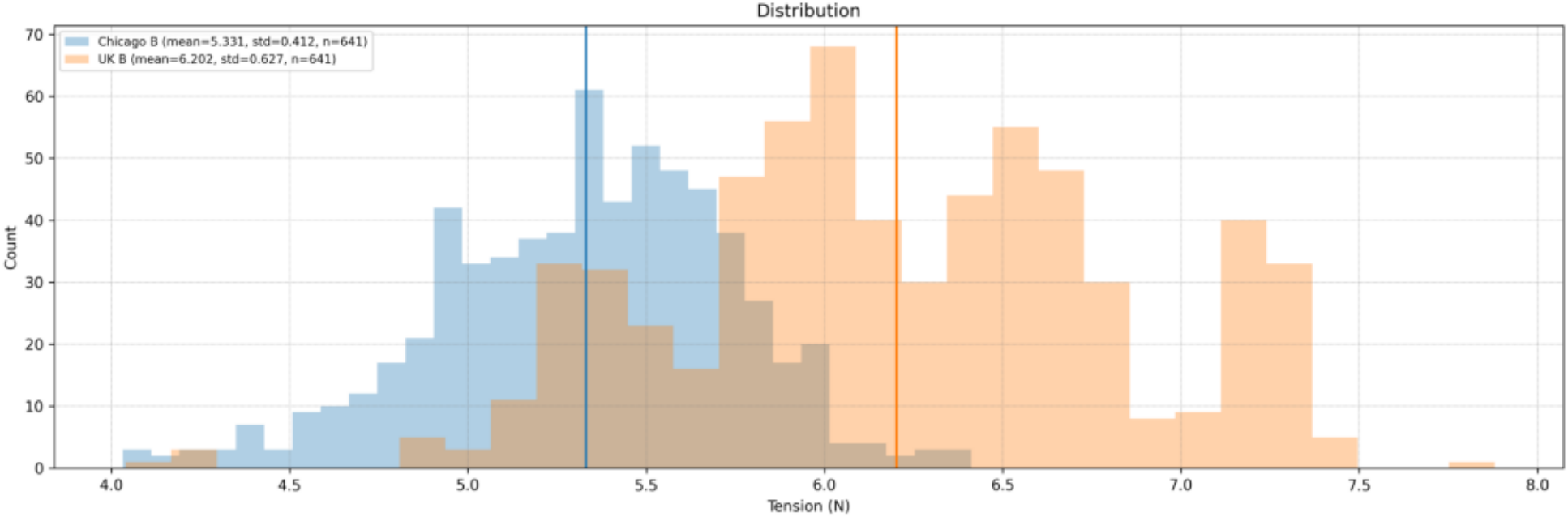
That shift improves the match, but only moderately. The comparison still shows an overall large negative offset between Chicago and UK measurements, and the fact that the U-layer coverage is still partial limits how far the interpretation can be pushed.

tension_raw_A_UKAPA7_U.png

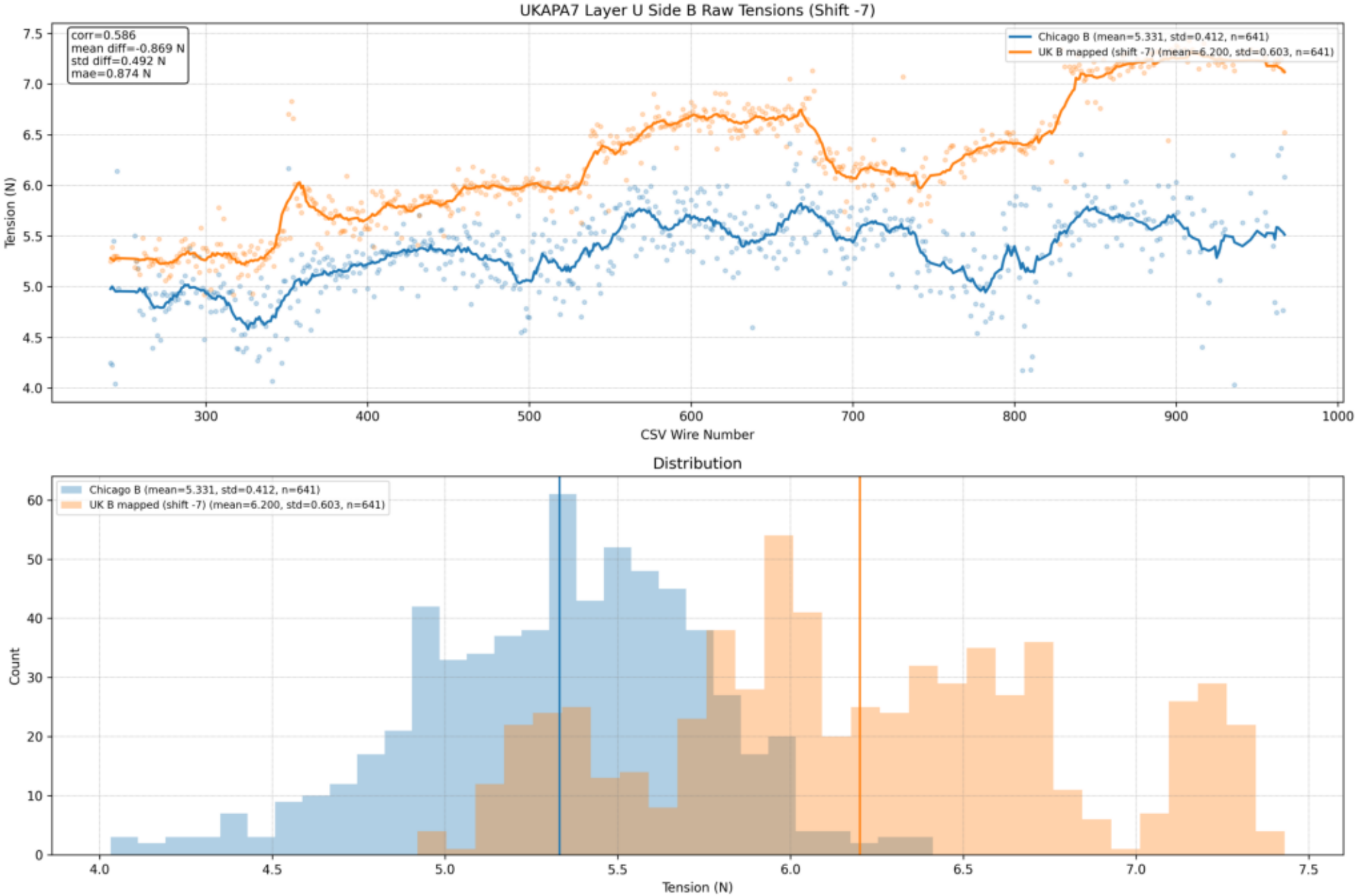


tension_raw_B_baseline_UKAPA7_U.png

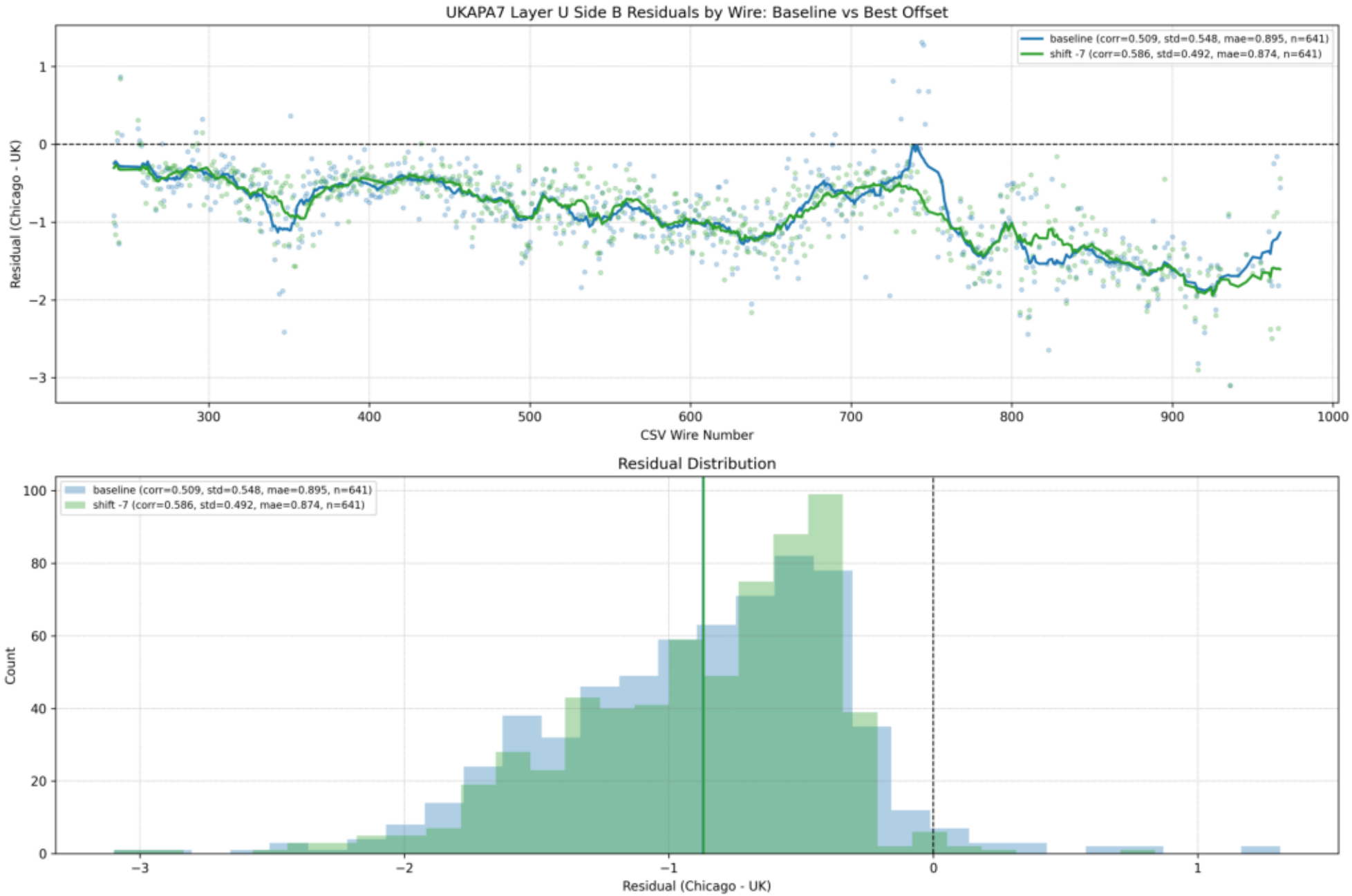
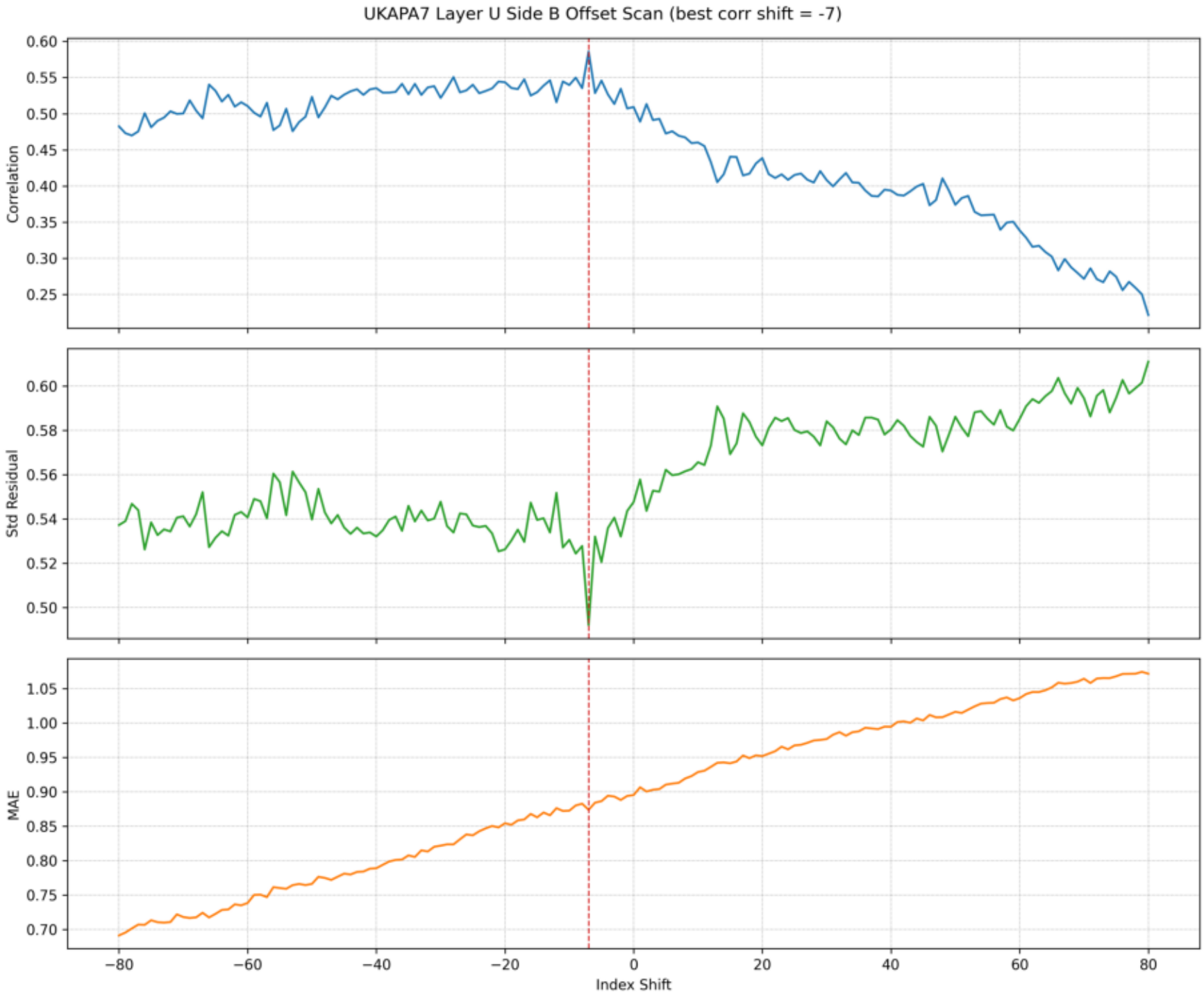




tension_raw_B_shifted_UKAPA7_U.png



tension_residual_B_offset_comparison_UKAPA7_U.png



tension_shift_scan_B_UKAPA7_U.png